

Software Engineering

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Interaction Diagram

Interaction diagrams show how different objects work together by sending messages to complete a specific task. In use cases, we have basic flows and alternate flows, and each flow can be shown with an interaction diagram. Use cases describe what happens (the behaviour), while sequence diagrams show how objects work together to make that behaviour happen.

For example, consider the “Withdraw Cash” scenario. The customer inserts a card into the ATM, and the ATM asks for a PIN. The customer enters the PIN, and the ATM checks if it is correct. Next, the ATM asks for the withdrawal amount. After the customer enters the amount, the ATM verifies that at least Rs. 500 will remain in the savings account after the withdrawal. If everything is fine, the ATM dispenses the cash, displays the remaining balance, and returns the card.

A class diagram shows the static structure of the system. It lists the classes, such as Customer, ATM, and Server, along with their attributes and operations, and shows how they are connected. However, a class diagram does not show the step-by-step actions during execution. For example, it does not explain which object communicates with which, or what happens after each step.

Interaction diagrams provide this detail. They show which object sends a message to which other object and what happens after each operation. UML provides two main types of interaction diagrams: sequence diagrams, which show the order of messages over time, and communication diagrams, which show the connections between objects and the messages they exchange.

As an example, consider the operation of an ATM card machine (simplified). The customer inserts a card in the machine. The machine prompts him for a personal identification number, i.e., pin code. The customer enters the pin code. The machine authenticates the pin code, displays the balance and asks the user to enter the amount that he wants to withdraw. After the user enters the amount, the machine verifies that the amount is less than Rs. 500 less than the amount present in the savings account of the customer. It then generates the cash and displays the balance in the customer's account. The customer ends the transaction and the machine outputs the card.

This scenario depicts the successful withdrawal of money by the customer. However, sometimes the procedure may not continue as depicted above.

1. When the customer types the pin, the system outputs that the pin is 'invalid' and asks the user to try again. This happens twice and if the pin is incorrect each time, the machine ends the transaction and outputs the card to the customer. If the pin was correct at any of the three attempts, then the all goes well scenario would have ensued.

2. The machine was informed that the inserted card was stolen. It confiscates the card and ends the transaction

3. If, after the successful validation of the pin code, the user enters an amount that leaves less than Rs. 500 in balance so that

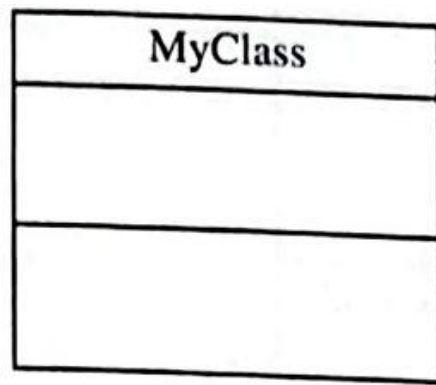
$$C \geq A \geq C - 500$$

where A=amount entered

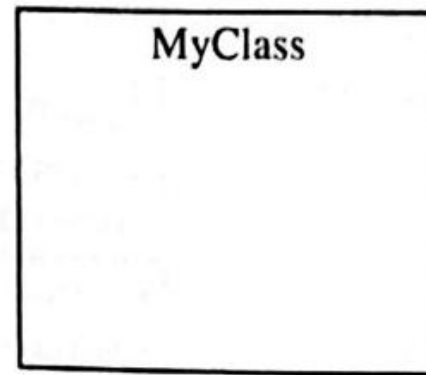
C=customer's total amount in savings account

then the machine outputs a warning about the fine that the customer would have to pay if his balance becomes less than Rs. 500, and asks him if he still wants to continue. If the user consents, then the machine outputs the cash, displays the balance in the customer's account, and shows him the fine that he has to pay within one month to the bank. The customer ends the transaction, and the machine outputs the card back to him

4. After the pin validation, the customer enters an amount greater than the amount in his savings account. The machine outputs the message and asks the user to re-enter. If he re-enters the correct amount, then the successful scenario will ensue, else if he enters an amount less than or equal to the amount in his savings account but which leaves less than Rs. 500 in his savings account, then case 3 will occur. If he persists in entering an amount greater than the amount in his savings account, the machine will give him two more opportunities and then output the card and end the transaction.



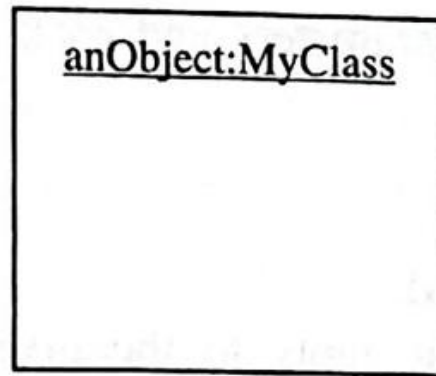
(a)



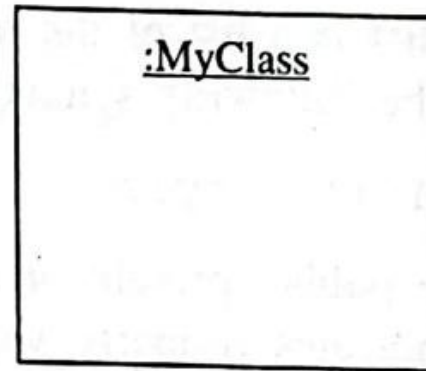
(b)

Figure 3.1 Two different representations of classes.

An object can be depicted as shown in Figure 3.2(a) and Figure 3.2(b). It is always underlined. The name of the object is followed by a colon and then by the class name from which it has been instantiated.

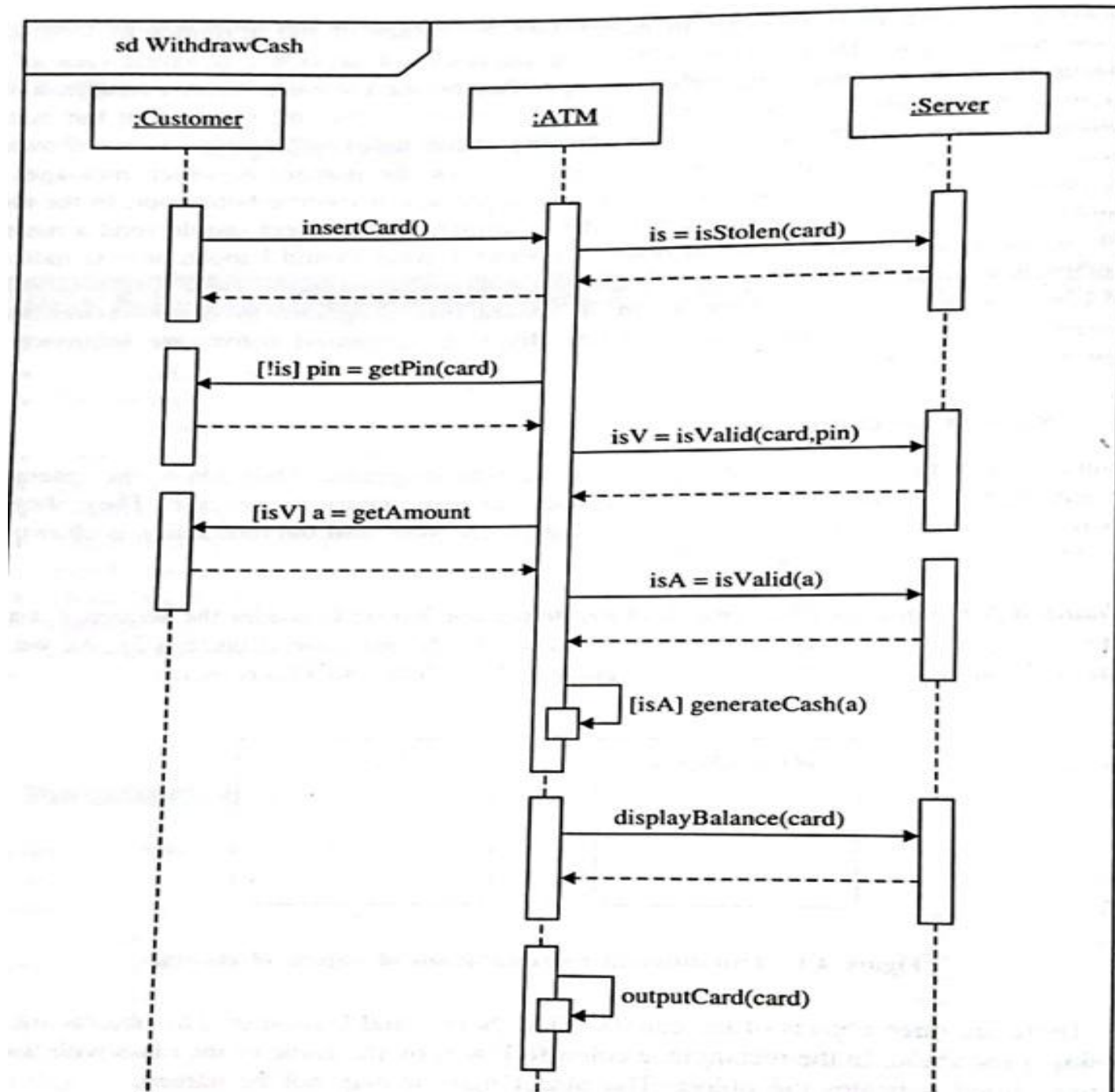


(a)

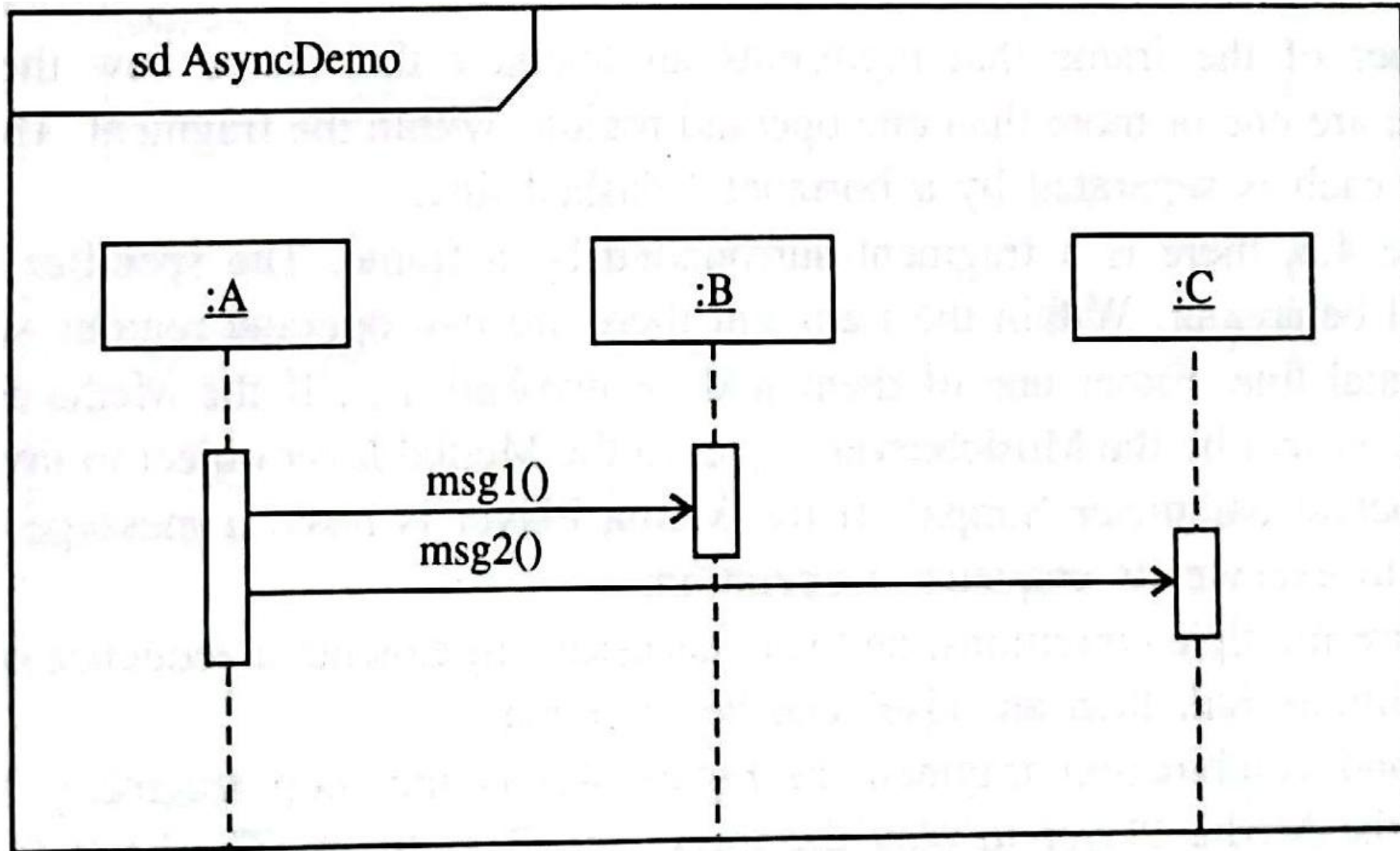


(b)

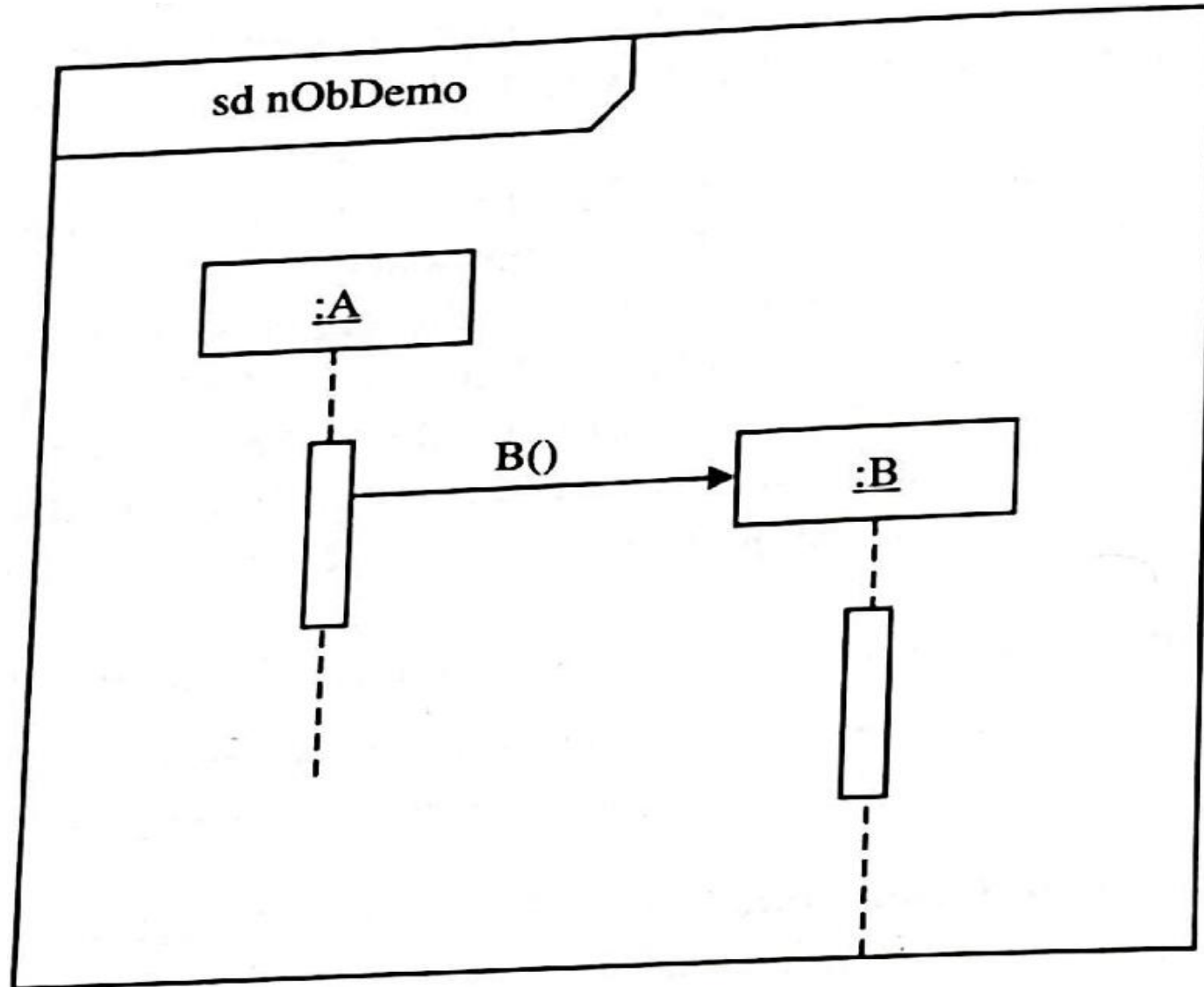
Figure 3.2 Two different representations of objects of classes.



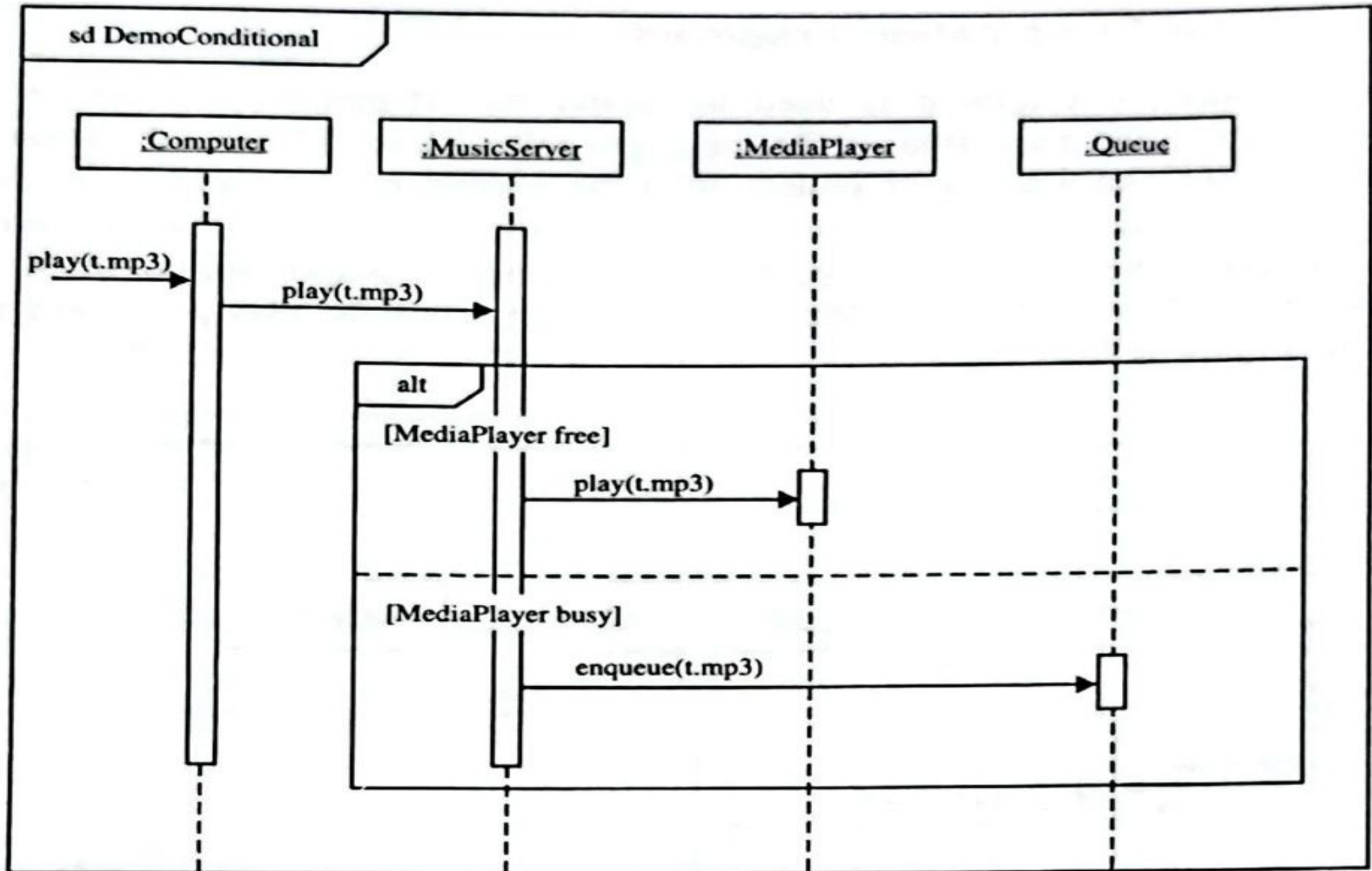
Asynchronous Messages

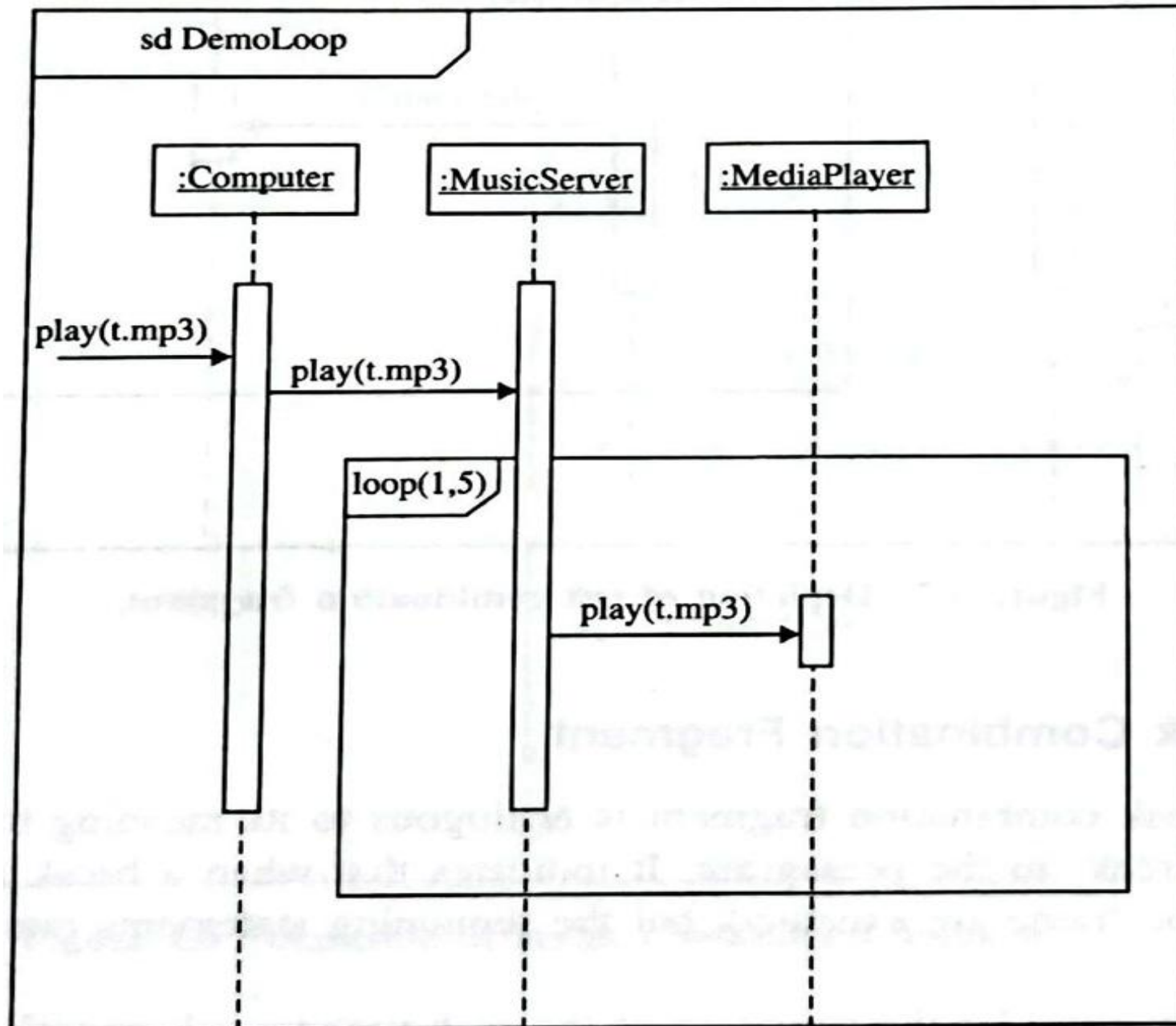


Creating New Objects

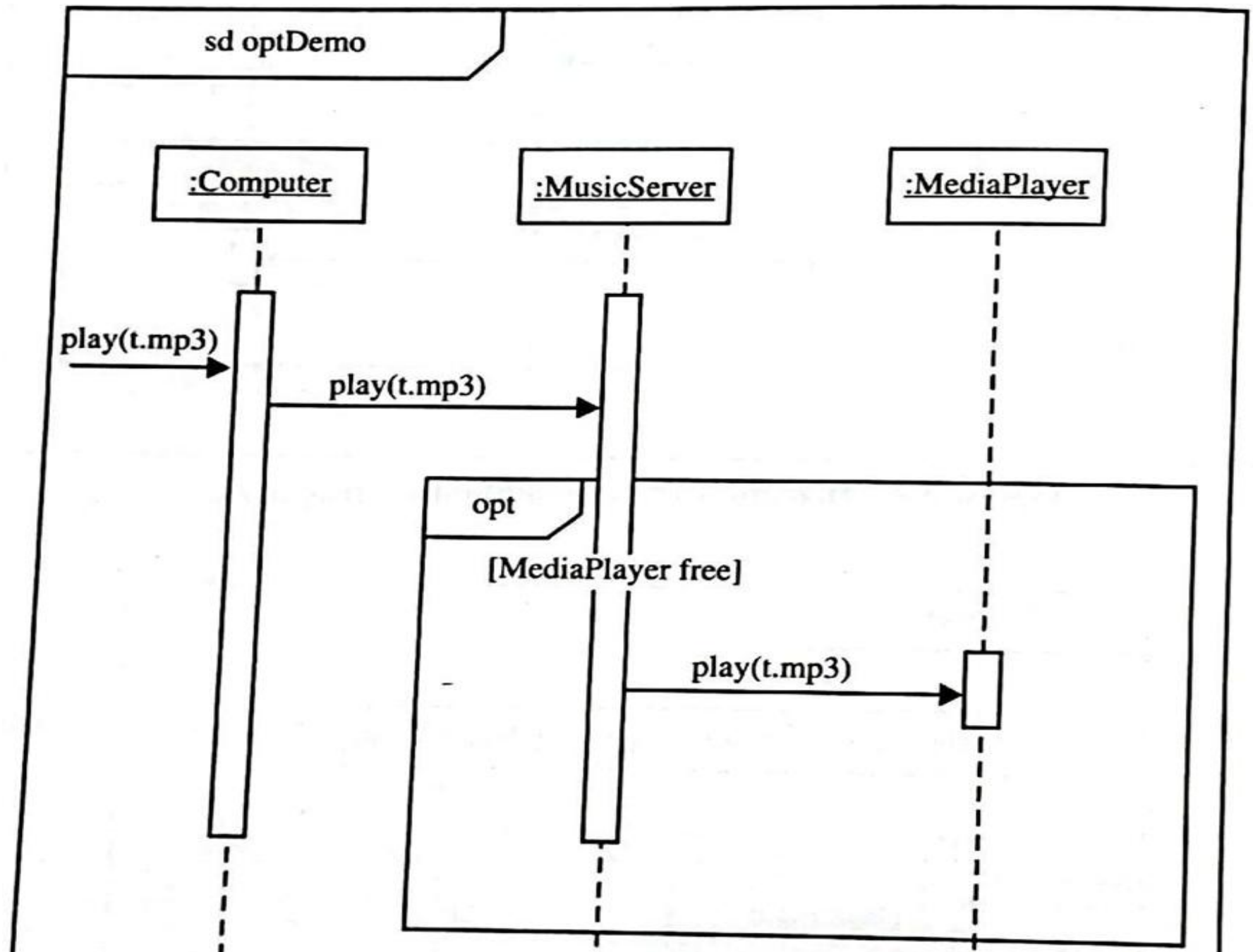


Condition and Loop

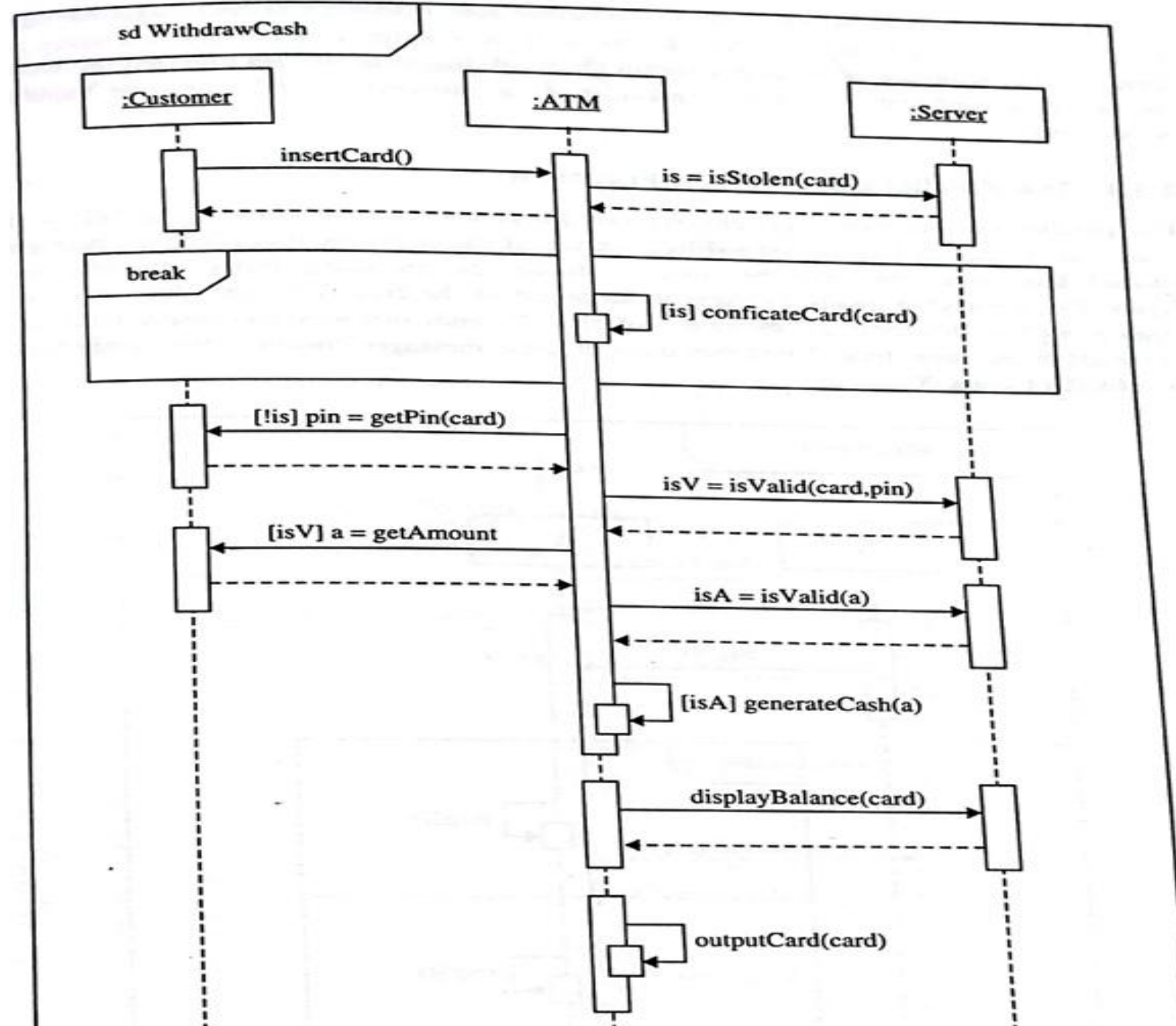




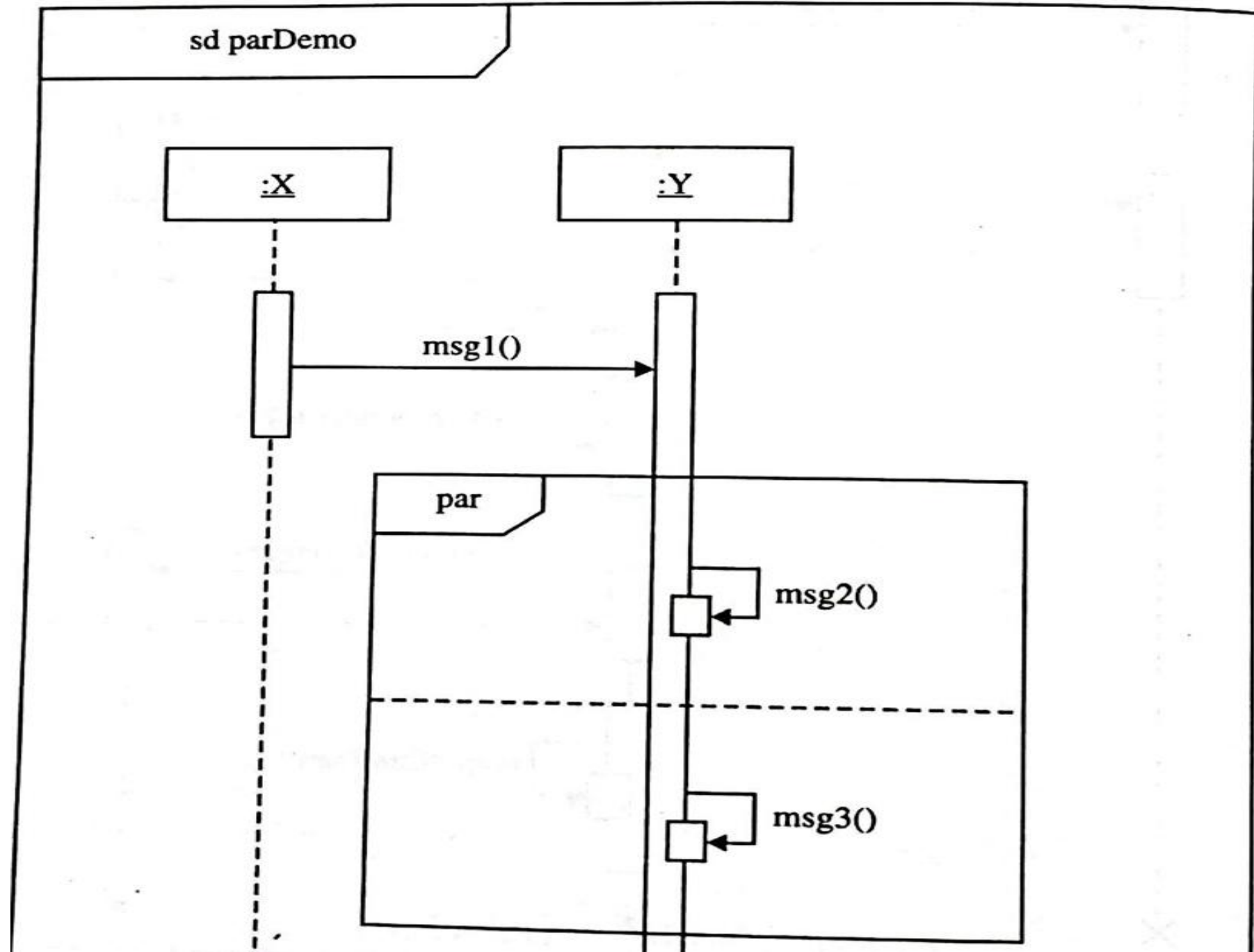
If-then Statement



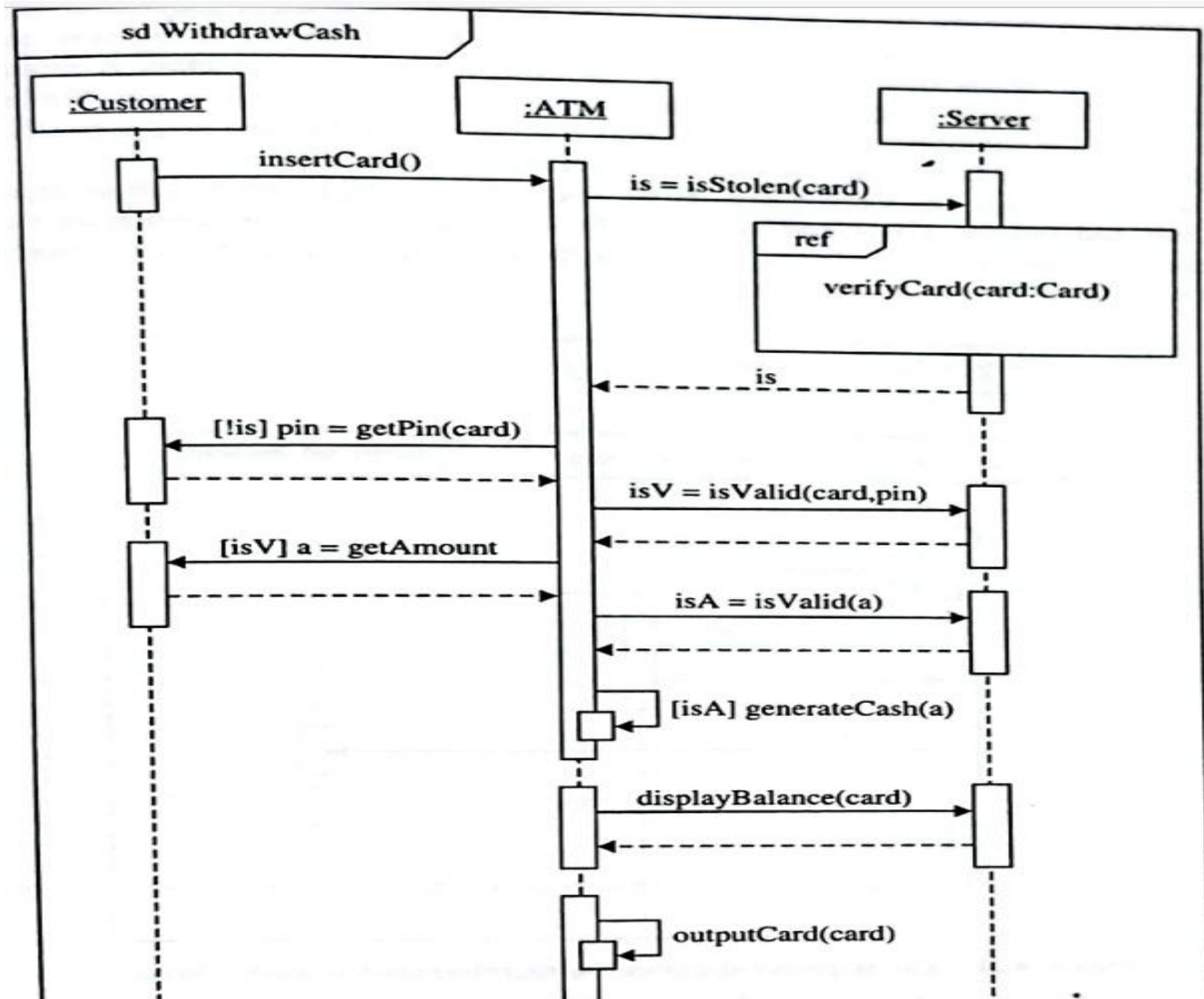
Break



Parallel



Ref



sd verifyCard(card:card):Boolean

:Server

:MainServer

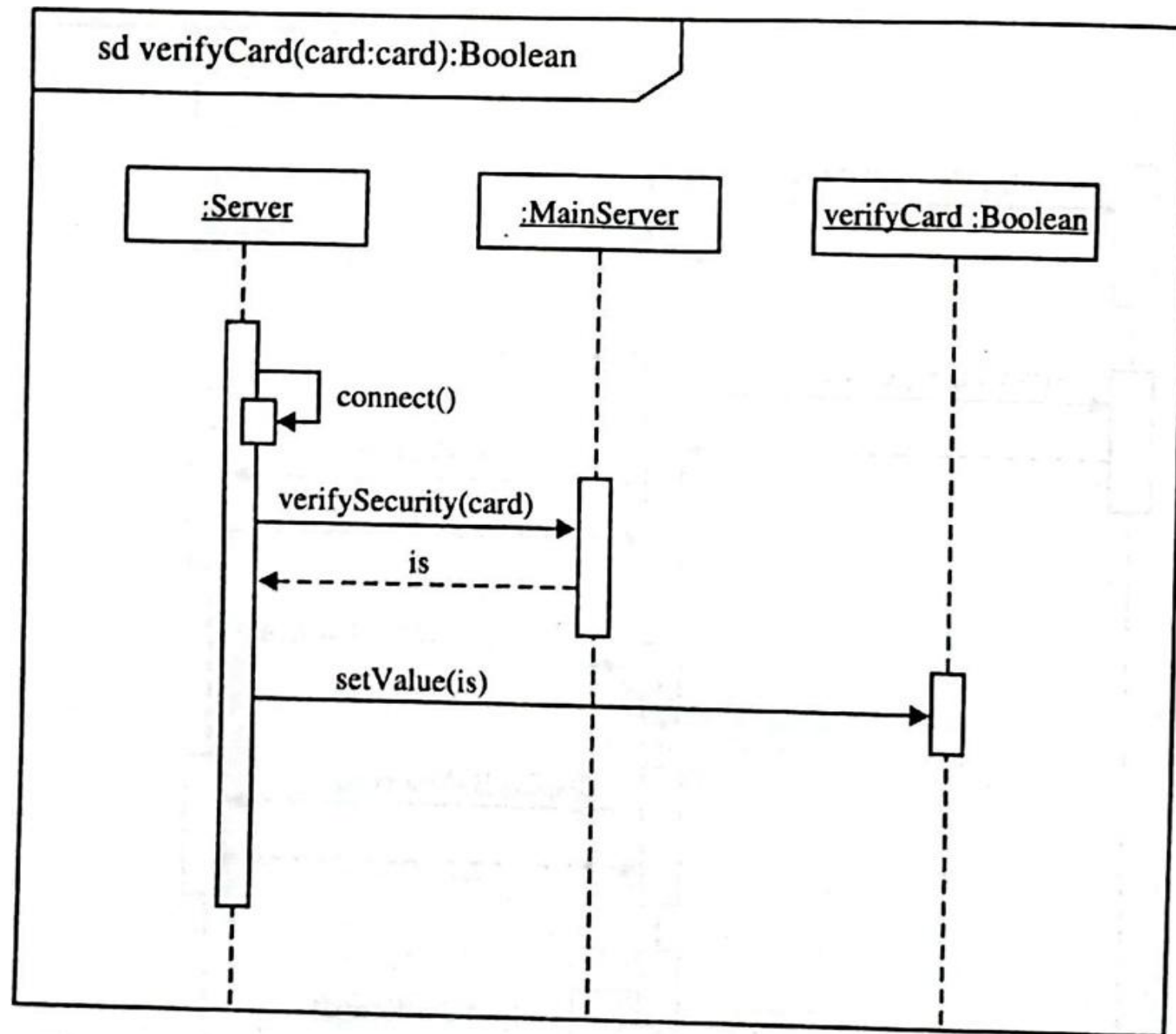
verifyCard :Boolean

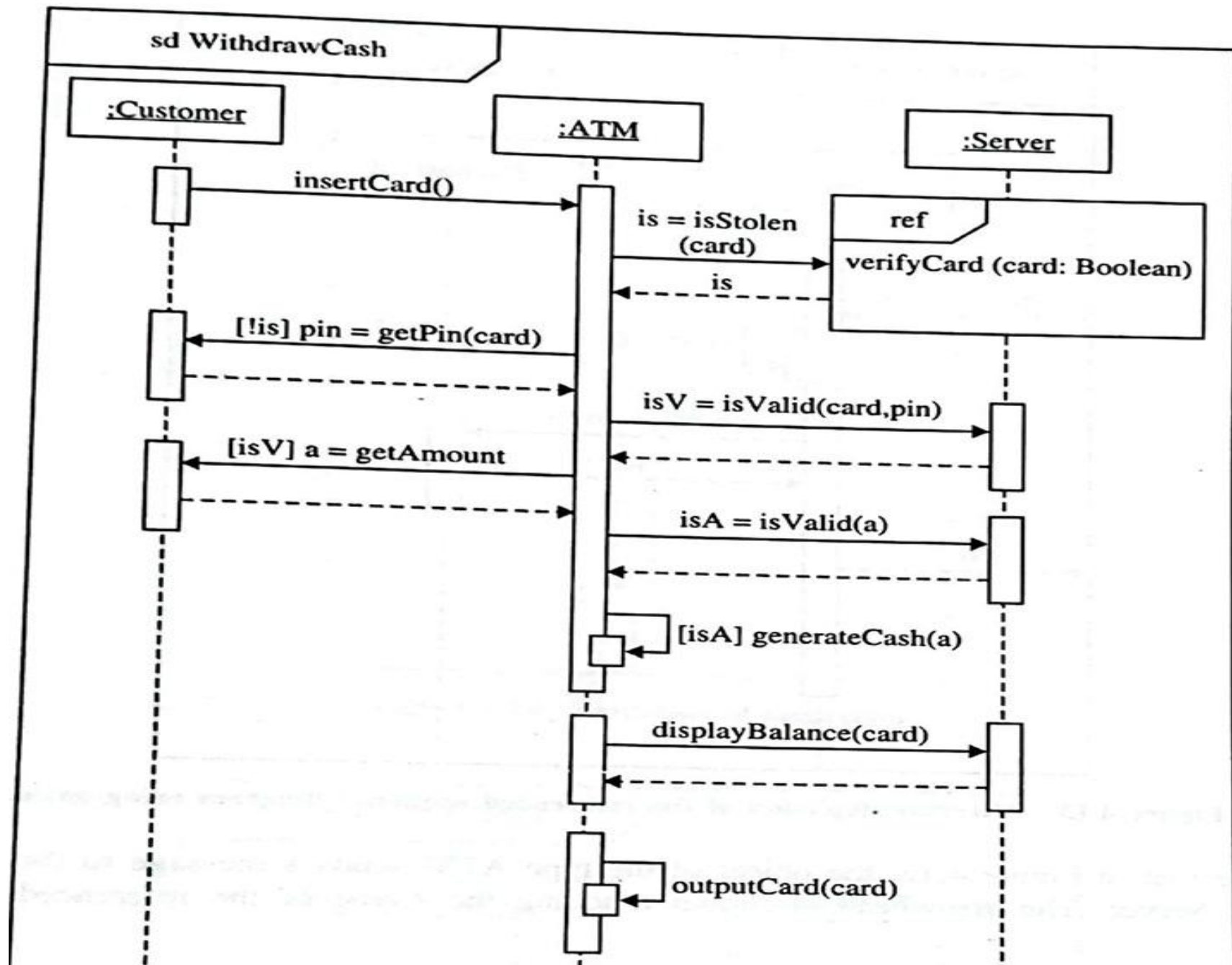
connect()

verifySecurity(card)

is

setValue(is)





Communication Diagram

Communication diagrams are another type of interaction diagram. They show how objects work together and send messages to each other to carry out a scenario in a use case. They also show how these objects are connected through communication paths.

When choosing between a sequence diagram and a communication diagram, it depends on what is more important for the modeller. If showing the order of events over time is important, use a sequence diagram. If showing the connections and relationships between objects is more important, use a communication diagram.

sd Music

play(t.mp3)

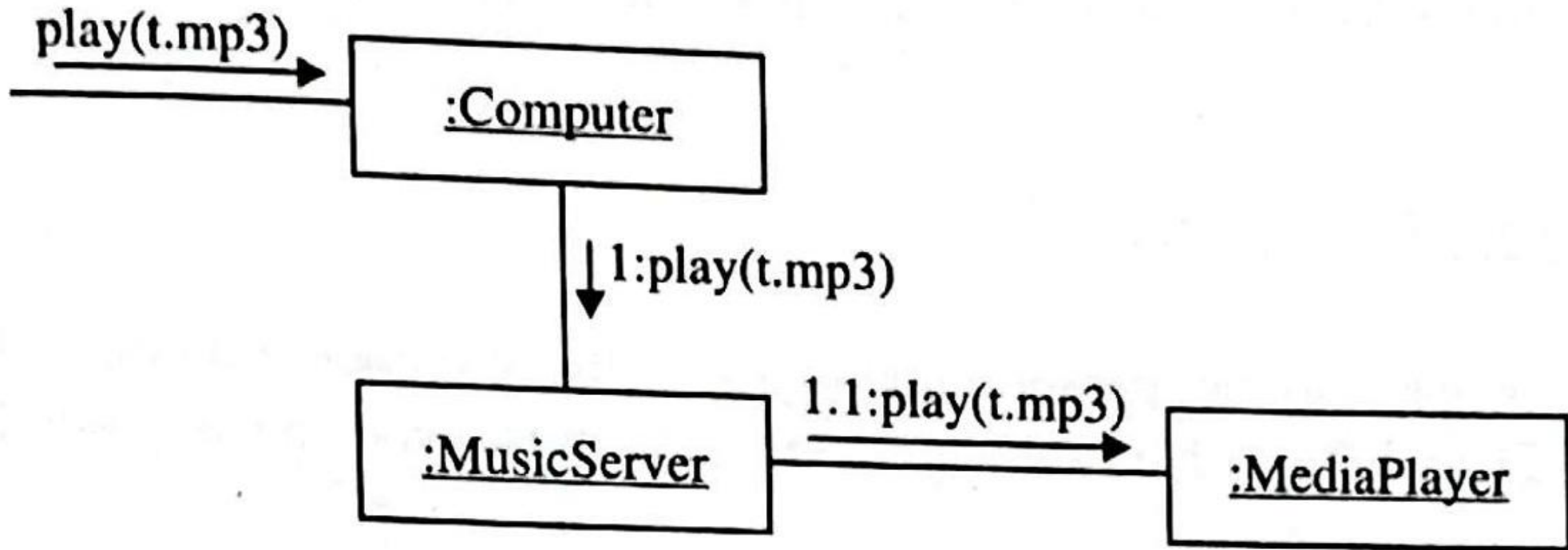
:Computer

↓ 1:play(t.mp3)

:MusicServer

1.1:play(t.mp3)

:MediaPlayer



How to draw Interaction Diagrams

- Step 1: Identify Interactions
- Step 2: Create Lifelines
- Step 3: Add Messages
- Step 4: Review and Refine
- Step 5: Validate

